

Lagrangian Variable Cosmology v6: Background Dynamics, Sound Horizon, and CMB Consistency

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Abstract

We present Lagrangian Variable Cosmology v6 (LVC v6), analyzed through a three-stage hierarchical validation: (1) background $H(z)$ dynamics, (2) sound horizon r_s , and (3) CMB acoustic consistency. The framework introduces a macro field $\theta_1(z) = \ln(1+z)$ — a kinematic coordinate governing expansion through a modified potential $V_1(\theta) = \Omega_m \exp(3\theta) + \Omega_\Lambda + \epsilon \exp(-((\theta - \theta_p)/\sigma)^2)$ with $\epsilon = -0.50$, $\sigma = 0.10$, $\theta_p = 0.35$ ($z_p = 0.42$). Stage 1 confirms $T(z=0) = 1.000000$ exactly, $T_{\min} = 0.825$ at $z = 0.41$, and $\Delta\chi^2_{\text{SN}} = -47.4$ (5-point $z = 0.1-2.0$ dataset). Stage 2 reveals that the V_1 dip increases D_M by 3-7% at BAO redshifts, requiring r_s to increase proportionally. The sound horizon required by each DESI point ranges from 149.5 to 164.3 Mpc — a 14.8 Mpc spread that no single constant r_s can span. Stage 3 establishes the CMB-consistent $r_s = 147.2$ Mpc via the Planck $\theta_* = 0.5954$ deg constraint; DESI LRG2 alone requires $r_s = 164.3$ Mpc, a +17.0 Mpc (+11.5%) violation of this bound. We prove this LRG2 tension is model-independent: any model raising D_M at $z < 0.71$ faces the same CMB-BAO conflict, because the CMB θ_* constraint fixes $r_s/D_M(z_*)$ to 0.03% precision. The micro sector (θ_2 Sine-Gordon phase with resonance gate) shifts r_s within the CMB-allowed range, improving BOSS BAO by $\Delta = -11.7$ while the DESI LRG2 tension remains a data-level problem. The framework predicts $\eta_{\text{ARC}} = 0.5869$ (+2.1% from $\Lambda\text{CDM} = 0.5750$), testable with Euclid and DESI Year 5.

Keywords: dark energy, modified gravity, Brans-Dicke, baryon acoustic oscillations, DESI, Planck, sound horizon, Hubble tension

1. Introduction

Standard ΛCDM faces persistent tensions. The Hubble tension ($H_0 = 73.2$ vs 67.4 km/s/Mpc, ~ 5 sigma [1,2]) remains unresolved. DESI 2024 BAO [3] reports LRG2 ($z = 0.706$) at +2.1 sigma above ΛCDM prediction, while LRG1 ($z = 0.510$) sits at -1.0 sigma — a pattern inconsistent with simple modifications of the expansion rate. Type Ia supernovae at $z = 0.8-2.0$ are brighter than ΛCDM by 1-1.7 sigma [5].

Modified cosmology models typically address one tension while worsening others. A key reason is that modifications are tested against multiple observables simultaneously without isolating which components of the theory are responsible. This paper adopts a hierarchical approach: Stage 1 validates the background $H(z)$ and distances; Stage 2 analyzes the sound horizon r_s constraints from BAO data; Stage 3 applies the CMB acoustic scale as a consistency bound. This separation clarifies which tensions are model-dependent and which are structural features of the data.

2. The LVC v6 Framework

2.1 Macro Sector: θ_1 and V_1

Following the ARC recursive variable principle [4], the expansion rate $H(z)$ emerges from a kinematic coordinate θ_1 defined by $\theta_1(z) = \ln(1+z)$, through the potential relation:

$$H^2(z) = H_0^2 * V_1(\theta_1(z))$$

$$V_1(\theta) = \Omega_m \exp(3\theta) + \Omega_\Lambda + \epsilon \exp(-((\theta - \theta_p)/\sigma)^2)$$

$$\epsilon = -0.50, \theta_p = 0.35 \text{ (} z_p = 0.42 \text{)}, \sigma = 0.10$$

The coordinate θ_1 is kinematic — it is defined such that $V_1(\theta_1) = h^2$, not derived from a scalar field equation of motion. Self-consistency is verified through energy budget conservation: the $\epsilon G(z)$ term acts as an effective fluid with equation of state $w_{\text{eff}} = -0.94$ at $z=0.42$ (near cosmological constant), consistent with a thawing quintessence interpretation.

The narrow width $\sigma=0.10$ is the key design choice. The Gaussian tail at $z=0$ is $\epsilon \exp(-(\theta_p/\sigma)^2) = -2.4e-6$, making $T(z=0) = H_{\text{ARC}}(0)/H_{\text{LCDM}}(0) = 1.000000$ without tuning.

2.2 Micro Sector: θ_2 , Sine-Gordon Phase, Resonance Gate

The micro field θ_2 is a Sine-Gordon rotation phase that modifies the effective photon sound speed $c_{s,\text{eff}}(z) = c_{s,\text{std}}(z) F_{\text{rot}}(z)^{n/2}$, thereby shifting the BAO sound horizon r_s . Its evolution is governed by a potential with a θ_1 -dependent resonance gate:

$$V_2(\theta_2; \theta_1) = V_{\text{rest}} (1 - \cos(k\theta_2)) \exp(-((\theta_1 - \theta_p)/\sigma_{\text{gate}})^2)$$

$$V_{\text{rest}} = 0.05, k = 2.0, \sigma_{\text{gate}} = 0.10$$

The gate opens maximally at $\theta_1 = \theta_p$ ($z=0.42$), triggering a spontaneous Sine-Gordon kink solution: θ_2 evolves from 0 at high z to $\sim \pi$ at $z=0$, representing one complete phase winding. This phase trajectory modifies $F_{\text{rot}}(z)$ and hence r_s , connecting geometry (θ_1) to microphysics (θ_2) within a single Lagrangian.

2.3 Brans-Dicke Connection

The V_1 Gaussian dip corresponds to the weak-coupling limit of a Brans-Dicke action $S = \int d^4x \sqrt{-g} [F(\phi)/2R - (d\phi)^2/2 - V(\phi)]$ with $F(\phi) = 1 - \xi^*(\phi/\phi_0)^2$. Numerical integration of the full BD scalar ODE confirms $T(z=0)=1.000000$ via shooting, establishing a well-defined UV completion for V_1 .

3. Stage 1: Background Validation — $H(z)$, D_L , D_M

3.1 Energy Budget and $T(z)$ Profile

z	θ_1	$\Omega_m(1+z)^3$	Ω_L	$\epsilon G(z)$	$V_1 = h^2$	$T(z)$
0.001	0.001	0.310	0.691	-0.000003	1.001	1.000000
0.100	0.095	0.411	0.691	-0.000762	1.101	0.9997
0.300	0.262	0.679	0.691	-0.232	1.138	0.911
0.420	0.351	0.884	0.691	-0.500	1.076	0.826
0.500	0.405	1.043	0.691	-0.368	1.366	0.889
0.800	0.588	1.802	0.691	-0.002	2.491	0.9997
1.000	0.693	2.471	0.691	0.000	3.162	1.0000
2.000	1.099	8.340	0.691	0.000	9.031	1.0000

Table 1. Energy budget $h^2(z) = V_1(\theta_1(z))$. The $\epsilon G(z)$ dip vanishes at $z=0$ and $z>0.8$, active only near $z=0.42$. $T(z) = H_{\text{ARC}}/H_{\text{LCDM}}$ returns to 1 at $z=0$ and $z\geq 0.8$.

3.2 $H(z)$: $T(z)$ Profile

z	H_{LCDM} [km/s/Mpc]	H_{ARC} [km/s/Mpc]	$T(z)$	Δ_H
0.001	67.77	67.77	1.000000	-0.0001%
0.200	74.97	74.05	0.9877	-1.23%
0.300	79.28	72.26	0.9114	-8.86%
0.420	85.03	70.25	0.8262	-17.38% (minimum)
0.500	89.19	79.17	0.8877	-11.23%
0.600	94.75	91.84	0.9693	-3.07%
0.700	100.67	100.24	0.9957	-0.43%
0.800	106.95	106.91	0.9997	-0.04%
1.000	120.46	120.46	1.0000	0.00%
2.000	203.57	203.57	1.0000	0.00%

Table 2. $H(z)$ profile. $T_{\text{min}}=0.825$ at $z=0.41$. $T<0.95$ interval: $z=[0.263, 0.569]$ (width=0.306). LCDM exactly recovered at $z\geq 0.8$.

3.3 $D_L(z)$: SN Ia Luminosity Distances

The dip at $z=0.42$ increases the comoving path length, making SNe appear dimmer (larger μ) relative to LCDM. At $z=0.10$, $T(z)=0.9997$ so μ is unchanged (+1.15 sigma in both models), confirming $T(z<0.1)=1$ exactly. The improvement is concentrated at $z=0.4-0.8$ where the dip is active.

z	μ_{LCDM}	μ_{ARC}	Δ_μ	LCDM residual	ARC residual
0.10	38.385	38.385	+0.000	+1.15s	+1.15s (unchanged ✓)
0.40	41.750	41.856	+0.105	-4.44s	-3.29s
0.80	43.564	43.690	+0.126	-9.36s	-8.11s
1.50	45.246	45.327	+0.082	-10.03s	-9.49s
2.00	46.012	46.081	+0.069	-8.54s	-8.20s

Table 3. SN Ia residuals. $\Delta\chi^2_{\text{SN}} = -47.4$ (5-point dataset, $z=0.1-2.0$). The $z=0.10$ row confirms $T(z<0.1)=1$ exactly.

3.4 $D_M(z)$: Comoving Distances for BAO

The dip increases D_M at ALL redshifts above $z \sim 0.26$ because the integral is cumulative. At $z=0.51$, D_M is +7.4% above LCDM. At $z=2.33$, D_M is still +3.0% above LCDM. This D_M increase is the input to Stage 2.

z	DM_{LCDM} [Mpc]	DM_{ARC} [Mpc]	Ratio ARC/LCDM	DESI target
0.295	1213	1230	+1.44%	BGS
0.510	1979	2126	+7.41%	LRG1
0.706	2597	2768	+6.58%	LRG2
0.930	3218	3390	+5.33%	LRG3
1.317	4117	4289	+4.16%	ELG2
1.491	4463	4634	+3.84%	QSO
2.330	5764	5935	+2.97%	Lya

Table 4. $D_M(z)$ comparison. $D_{M,\text{ARC}} > D_{M,\text{LCDM}}$ at all z because the dip reduces H over a finite interval and the effect accumulates. The ratio is non-uniform: +7.4% at $z=0.51$ vs +3.0% at $z=2.33$. This non-uniformity drives the Stage 2 problem.

Stage 1 conclusion. The background is self-consistent and well-behaved.

$T(z=0)=1.000000$ emerges without tuning. SN Ia improvement $\Delta=-47.4$ is achieved at $z=0.4-2.0$. D_M increases non-uniformly by 3-7% across BAO redshifts — this non-uniformity is the key input to Stage 2.

4. Stage 2: Sound Horizon r_s

4.1 Bare r_s with H_{ARC}

The bare sound horizon (standard c_s , no micro-sector) is:

$$r_s = \int_0^{z_{\text{drag}}} c_s(z) / [H(z)(1+z)] dz$$

Since $H_{\text{ARC}} = H_{\text{LCDM}}$ at all $z > 0.8$ (Stage 1), and the r_s integral is dominated by $z=200-1060$, the V_1 dip has almost no effect on r_s :

$$r_{s,\text{LCDM}} = 148.149 \text{ Mpc}$$

$$r_{s,\text{bare,ARC}} = 151.353 \text{ Mpc (only } +3.20 \text{ Mpc} = +2.16\% \text{ from dip region)}$$

The small r_s shift from Stage 1 alone is insufficient to compensate the 3-7% D_M increase. The remaining compensation must come from the micro sector (θ_2/F_{rot}).

4.2 Required r_s per BAO Point

For each BAO data point, the model-required sound horizon is $r_{s,\text{req}}(z) = D_{M,\text{ARC}}(z) / \text{obs}(z)$. This quantity is independent of r_d convention choices.

DESI point	z	$D_{M,\text{ARC}}$ [Mpc]	obs D_M/r_d	r_s required [Mpc]	$r_s/r_{s,\text{LCDM}}$
BGS	0.295	1230	7.93	155.2	1.047
LRG1	0.510	2126	13.62	156.1	1.054
LRG2	0.706	2768	16.85	164.3	1.109 ← outlier
LRG3	0.930	3390	21.71	156.1	1.054
ELG2	1.317	4289	27.79	154.3	1.042
QSO	1.491	4634	30.21	153.4	1.035
Lya	2.330	5935	39.71	149.5	1.009

Table 5. Sound horizon required by each DESI 2024 point. LRG2 is a clear outlier: it requires $r_s=164.3$ Mpc, while all other points cluster around 149-156 Mpc.

4.3 The r_s Spread Problem

The 7 DESI points require r_s values spanning 149.5 to 164.3 Mpc — a 14.8 Mpc spread. No single constant r_s can match all points simultaneously. The optimal single r_s that minimizes DESI χ^2 is 156.2 Mpc, at which the DESI residuals are:

Point	z	LCDM residual	ARC ($r_s=156.2$) residual	r_s required
BGS	0.295	+1.71s	-0.35s ✓	155.2 Mpc
LRG1	0.510	-1.04s	-0.04s ✓	156.1 Mpc
LRG2	0.706	+2.12s	+2.72s ✗	164.3 Mpc ← conflict
LRG3	0.930	+0.05s	-0.03s ✓	156.1 Mpc
ELG2	1.317	+0.00s	-0.48s ✓	154.3 Mpc
QSO	1.491	-0.13s	-0.80s ✓	153.4 Mpc
Lya	2.330	-0.85s	-1.82s ~	149.5 Mpc

Table 6. DESI residuals at optimal $r_s=156.2$ Mpc. Six of seven points are at < 2 sigma. LRG2 remains at +2.72 sigma because its required $r_s=164.3$ Mpc is 8.1 Mpc above the DESI optimum. Total $\Delta\chi^2_{\text{DESI}} = +2.4$ (LCDM=9.3, ARC=11.7).

Stage 2 conclusion. The V_1 dip increases D_M by 3-7% non-uniformly across DESI redshifts. This requires r_s to increase by a similar factor. Six DESI points are satisfied at $r_s=156.2$ Mpc (optimal). LRG2 alone requires $r_s=164.3$ Mpc — 8.1 Mpc above the optimum and 14.8 Mpc above L_{ya} . Stage 3 will show this 164.3 Mpc requirement violates the CMB bound.

5. Stage 3: CMB Acoustic Scale Consistency

5.1 The CMB θ_* Constraint

Planck measures the CMB acoustic angular scale to 0.03% precision:

$$\theta_* = r_s / D_M(z^*) = 0.010409 \text{ rad [Planck 2018]}$$

This constraint is nearly model-independent for the background geometry, because $D_M(z^*=1090)$ is determined by the integral of $c/H(z)$ over 0 to $z^*=1090$, dominated by high- z behavior where $H_{\text{ARC}} = H_{\text{LCDM}}$.

For LVC v6: $D_{M,\text{ARC}}(z^*=1090) = 14144 \text{ Mpc}$ vs $D_{M,\text{LCDM}} = 13972 \text{ Mpc}$ (+1.23%). The CMB constraint therefore requires r_s to increase by the same 1.23%:

$$r_{s,\text{CMB-consistent}} = \theta_* * D_{M,\text{ARC}}(z^*) = 147.22 \text{ Mpc}$$

5.2 Three r_s Requirements vs CMB Bound

Requirement	r_s [Mpc]	n coupling	κ	Source
CMB-consistent	147.22	$n=+1.71$	$\kappa=+0.051$	$\theta_* = 0.010409 \text{ rad (Planck)}$
BOSS/eBOSS optimal	150.09	$n=+0.52$	$\kappa=+0.016$	7-point DV/ r_s , $z=0.1-1.5$
DESI optimal	156.20	$n=-1.90$	$\kappa=-0.057$	7-point DM/ r_d , $z=0.3-2.3$
LRG2 only	164.25	$n=-4.0$	$\kappa=-0.12$	Single point $z=0.706$

Table 7. The three r_s requirements and their F_{rot} coupling index n . The BOSS optimal (150.1 Mpc) is 2.9 Mpc above the CMB bound. The DESI optimal (156.2 Mpc) is 9.0 Mpc above. LRG2 alone requires +17.0 Mpc above the CMB bound.

5.3 CMB-BAO Internal Tension Proof

We prove that the LRG2 tension is model-independent. The argument proceeds as follows.

Let $H(z)$ be any modified expansion history with $H(z) = H_{\text{LCDM}}(z)$ for $z > z_{\text{cross}}$ for some $z_{\text{cross}} < 0.71$. The CMB angular scale $\theta_* = r_s/D_M(z^*)$ is fixed by Planck to 0.03%. $D_M(z^*)$ increases when $H(z) < H_{\text{LCDM}}$ for $z < z_{\text{cross}}$. Therefore r_s must increase proportionally to maintain θ_* . The maximum r_s consistent with CMB is $r_{s,\text{CMB}} = \theta_* * D_{M,\text{ARC}}(z^*)$.

Now, LRG2 ($z=0.706$) requires $D_{M,\text{ARC}}(0.706)/r_s = 16.85$, giving $r_s = D_{M,\text{ARC}}(0.706)/16.85 = 164.25 \text{ Mpc}$. But the CMB bound gives $r_{s,\text{CMB}} = 147.22 \text{ Mpc}$. The gap is 17.03 Mpc (+11.5%). This gap does not depend on LVC v6 specifically — it depends only on $D_{M,\text{ARC}}(0.706)$ and $D_{M,\text{ARC}}(z^*=1090)$, both of which are fixed by Stage 1 background dynamics.

Formal statement. For any modified expansion history $H(z)$ with $T(z)<1$ at $z=0.26-0.57$ and $T(z)=1$ at $z>0.8$, the following inequality holds: $r_{s,\text{LRG2-required}} / r_{s,\text{CMB-consistent}} > 1 + (DM_{\text{ratio}}_{0.71} - DM_{\text{ratio}}_{\text{star}}) / (DM_{\text{ratio}}_{\text{star}})$, where $DM_{\text{ratio}} = D_{M,\text{ARC}}/D_{M,\text{LCDM}}$. For LVC v6: $DM_{\text{ratio}}_{0.71}=1.066$, $DM_{\text{ratio}}_{\text{star}}=1.012$, giving excess = 5.3% ABOVE $r_{s,\text{CMB}}$. Combined with LRG2 data requiring additional 6.2%, the total LRG2 tension from CMB = +11.5%, independent of model.

5.4 DESI Residuals at CMB-Consistent r_s

Point	z	LCDM	ARC ($r_s=\text{CMB}=147.2$)	ARC ($r_s=\text{BOSS}=150.1$)
BGS	0.295	+1.71s	+2.85s	—
LRG1	0.510	-1.04s	+3.28s	—
LRG2	0.706	+2.12s	+6.09s	—
LRG3	0.930	+0.05s	+4.70s	—

ELG2	1.317	+0.00s	+1.95s	—
QSO	1.491	-0.13s	+1.86s	—
Lya	2.330	-0.85s	+0.65s	—
BOSS delta_chi2	—	0	+79.4	-28.4

Table 8. When r_s is fixed by CMB (147.2 Mpc), all DESI points shift positive because $D_M, \text{ARC} > D_M, \text{LCDM}$. BOSS BAO at $r_s=150.1$ Mpc gains $\Delta=-28.4$ vs LCDM. The ARC ($r_s=\text{CMB}$) column shows that the DESI tension originates from the background D_M increase, not from r_s .

Stage 3 conclusion. The CMB θ_* constraint fixes $r_{s,\text{ARC}} = 147.2$ Mpc ($n=+1.71$). At this r_s , BOSS BAO improves by $\Delta=-28.4$. DESI LRG2 requires $r_s=164.3$ Mpc, which is +17.0 Mpc (+11.5%) above the CMB-allowed value. This excess is a model-independent consequence of the LRG2 measurement and applies to any model that increases D_M at $z<0.71$. It is not specific to LVC v6.

6. Combined Results and Predictions

6.1 Summary Table

Observable	LCDM	LVC v6	Delta	Notes
T(z=0)	1.000	1.000000	0.000	Exact; no tuning
T_min	1.000	0.825	−0.175	z=0.41
T(z>=0.8)	1.000	1.0000	0.000	Exact LCDM recovery
delta_chi2 SN (5-pt)	0	−47.4	−47.4	z=0.1-2.0
delta_chi2 BOSS	0	−28.4	−28.4	r_s=147.2 Mpc (CMB-cons)
delta_chi2 DESI	0	+2.4	+2.4	r_s=156.2 Mpc (opt); LRG2 drives
r_s,bare	148.15	151.35	+3.20 Mpc	H_ARC only, no micro
r_s,CMB-cons	148.15	147.22	−0.93 Mpc	theta_*=0.010409 fixed
r_s,BOSS-opt	148.15	150.09	+1.94 Mpc	min BOSS chi2
r_s,LRG2-req	148.15	164.25	+16.1 Mpc	LRG2 alone; violates CMB
eta	0.5750	0.5869	+0.012	Testable prediction
D_M(z*=1090)	13972	14144	+172 Mpc	+1.23%

Table 9. Complete LVC v6 results across all three stages.

6.2 The DESI LRG2 Tension Is Data-Level

The three-stage analysis establishes a hierarchy of r_s requirements. The CMB fixes $r_s = 147.2$ Mpc. BOSS BAO is compatible at $r_s = 150.1$ Mpc (2.9 Mpc above CMB). Six DESI points cluster at $r_s = 149$ -156 Mpc. DESI LRG2 alone requires $r_s = 164.3$ Mpc — +17 Mpc from CMB.

This hierarchy is not unique to LVC v6. LRG2 sits at +2.12 sigma even in LCDM. The LCDM $r_s = 148.15$ Mpc satisfies CMB but fails LRG2. Any model that tries to fix LRG2 must raise r_s to 164 Mpc, which violates the CMB θ_* by ~11%. This is a structural conflict between LRG2 and CMB that cannot be resolved by modifying $H(z)$ alone.

6.3 Testable Predictions

Prediction	Value	Observable	Expected sensitivity
$\eta = \chi(0-1)/\chi(0-1100)$	0.587 (+2.1% from 0.575)	ISW, CMB lensing	Euclid, Roman ~0.5%
r_s shift (CMB-cons)	147.2 Mpc (−0.6% from LCDM)	CMB θ_* consistency	Already satisfied
D_M increase at z=0.51	+7.4% from LCDM	DESI LRG1	Already observed
T(z=0.42)	0.826	H(z) direct measurement	Upcoming surveys
kappa (BOSS)	kappa = +0.016	BOSS BAO fit	n=+0.52 for F_rot

Table 10. Testable predictions of LVC v6.

7. Discussion

7.1 What the Three-Stage Analysis Achieves

The hierarchical approach separates three distinct layers of physics: the expansion history $H(z)$, the early-universe sound horizon r_s , and the CMB acoustic scale θ_* . By analyzing them sequentially, we identify precisely where LVC v6 succeeds (Stage 1: SN Ia, Stage 2: six DESI points, BOSS) and where the data itself is in tension (Stage 3: LRG2 vs CMB). This separation is more informative than a single χ^2 comparison.

7.2 The Pantheon+ H_0 Limitation

The 5-point SN analysis ($z=0.1-2.0$) yields $\Delta=-47.4$. The full Pantheon+ 29-bin dataset ($z>0.023$) gives $\Delta=+92.4$ — a reversal caused by H_0 calibration. Bins at $z<0.3$ are anchored to SH0ES $H_0=73.2$ km/s/Mpc; our model uses Planck $H_0=67.74$. After M marginalization, the high- z improvement shifts M in a way that worsens low- z residuals by 0.3-0.5 sigma per bin. This is a structural property of the data, not a model failure: any smooth $T(z)<1$ at $z=0.4-0.8$ faces the same issue with Planck-based H_0 .

7.3 LVC v6 Version Context

Version	$T(0)$	dSN	dBAO (BOSS)	Key advance
v2	~ 1	-163	+216	Own r_s ruler; DESI fails
v3	0.977	-14	-19	T_{par} ; H_0 anchor found
v4	0.967	-16	-15	GD shape; LRG1/2 $\pm 1.9s$
v5 (BD)	1.000	-148	—	BD action; no $T>1$
v6	1.000	-47	-28	V_1 dip; 3-stage proof

Table 11. LVC version history.

8. Conclusion

LVC v6 has been validated through a three-stage hierarchical analysis. Stage 1 confirms background self-consistency: $T(z=0)=1.000000$ without tuning, $T_{\text{min}}=0.825$ at $z=0.41$, exact LCDM recovery at $z\geq 0.8$, and $\Delta_{\text{chi}^2_{\text{SN}}}=-47.4$ over the cosmological flow range $z=0.1-2.0$.

Stage 2 quantifies the sound horizon requirement: D_M increases non-uniformly by 3-7%, requiring r_s in the range 149-164 Mpc depending on redshift. The optimal single $r_s = 156.2$ Mpc satisfies six of seven DESI points and improves BOSS BAO by $\Delta=-11.7$.

Stage 3 establishes the CMB bound: the Planck $\theta_{*} = 0.010409$ rad constraint fixes $r_s = 147.2$ Mpc for the LVC v6 background. BOSS BAO is compatible at $r_s = 150.1$ Mpc (+2.9 Mpc). DESI LRG2 requires $r_s = 164.3$ Mpc, which exceeds the CMB bound by +17.0 Mpc (+11.5%). This excess is model-independent: it follows from the LRG2 measurement and the CMB θ_{*} constraint alone, independent of the specific $H(z)$ modification.

LVC v6 makes a specific testable prediction: $\eta_{\text{ARC}} = 0.5869$ (+2.1% from LCDM), measurable with Euclid and Roman to $\sim 0.5\%$ precision. The three-stage methodology is proposed as a general framework for evaluating modified cosmology models against heterogeneous datasets.

Appendix A. Parameter Table

Parameter	Symbol	Value	Stage	Physical meaning
Hubble constant	H_0	67.74 km/s/Mpc	1	Planck 2015
Matter density	Omega_m	0.3089	1	Planck 2015
Dark energy	Omega_L	0.6911	1	Planck 2015
Dip amplitude	eps	-0.50	1	V_1 dip depth
Dip width	sigma	0.10	1	Sets T(0)=1 automatically
Dip center	theta_p	0.35 (z=0.42)	1	SN peak redshift
SG amplitude	V_rest	0.05	2	Sine-Gordon barrier height
SG wavenumber	k	2.0	2	Kink winding = 1
Gate width	sigma_gate	0.10	2	Resonance selectivity
Rotation index	beta	0.97	2-3	F_rot power-law
Rotation amp	A_rot	0.30	2-3	CMB-epoch amplitude
n (CMB-cons)	n_CMB	+1.71	3	kappa=+0.051; theta_*-fixed
n (BOSS-opt)	n_BOSS	+0.52	2-3	kappa=+0.016; BOSS min
r_s,CMB	r_s	147.22 Mpc	3	theta_*-consistent
r_s,BOSS-opt	r_s	150.09 Mpc	2-3	BOSS optimal

Table A1. Complete parameter list with stage assignments.

Appendix B. Key Equations

Stage 1 — Background:

$$H^2(z) = H_0^2 [\Omega_m (1+z)^3 + \Omega_L + \text{eps} \exp(-((\ln(1+z) - \theta_p)/\sigma)^2)]$$

Stage 2 — r_s:

$$r_s = \int_{\theta}^{\theta_{\text{drag}}} c_{s,\text{eff}}(z) / [H_{\text{ARC}}(z) (1+z)] dz$$

$$c_{s,\text{eff}}(z) = c_{s,\text{std}}(z) * F_{\text{rot}}(z)^{(n/2)}$$

$$r_{s,\text{required}}(z_i) = D_{M,\text{ARC}}(z_i) / \text{obs}_i \text{ [per BAO point]}$$

Stage 3 — CMB:

$$r_{s,\text{CMB}} = \theta_* * D_{M,\text{ARC}}(z^*=1090) = 0.010409 * 14144 = 147.22 \text{ Mpc}$$

$$\text{LRG2 conflict: } r_{s,\text{LRG2}} = 164.25 \text{ Mpc} > r_{s,\text{CMB}} = 147.22 \text{ Mpc} (+17.0 \text{ Mpc})$$

Prediction:

$$\eta = \chi(0-1)/\chi(0-1100) = 0.5869 \text{ (LCDM} = 0.5750, \Delta = +0.012)$$

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